

● HARD

● CRAFT AND STRUCTURE

Cross-Text Connections

10 questions · ~15 min

01

Text 1

Growth in the use of novel nanohybrids—materials created from the conjugation of multiple distinct nanomaterials, such as iron oxide and gold nanomaterials conjugated for use in magnetic imaging—has outpaced studies of nanohybrids’ environmental risks. Unfortunately, risk evaluations based on nanohybrids’ constituents are not reliable: conjugation may alter constituents’ physiochemical properties such that innocuous nanomaterials form a nanohybrid that is anything but.

Text 2

The potential for enhanced toxicity of nanohybrids relative to the toxicity of constituent nanomaterials has drawn deserved attention, but the effects of nanomaterial conjugation vary by case. For instance, it was recently shown that a nanohybrid of silicon dioxide and zinc oxide preserved the desired optical transparency of zinc oxide nanoparticles while mitigating the nanoparticles’ potential to damage DNA.

Based on the texts, how would the author of Text 2 most likely respond to the assertion in the underlined portion of Text 1?

- A. By concurring that the risk described in Text 1 should be evaluated but emphasizing that the risk is more than offset by the potential benefits of nanomaterial conjugation
- B. By arguing that the situation described in Text 1 may not be representative but conceding that the effects of nanomaterial conjugation are harder to predict than researchers had expected
- C. By denying that the circumstance described in Text 1 is likely to occur but acknowledging that many aspects of nanomaterial conjugation are still poorly understood

- D. By agreeing that the possibility described in Text 1 is a cause for concern but pointing out that nanomaterial conjugation does not inevitably produce that result

Text 1

Despite its beautiful prose, *The Guns of August*, Barbara Tuchman's 1962 analysis of the start of World War I, has certain weaknesses as a work of history. It fails to address events in Eastern Europe just before the outbreak of hostilities, thereby giving the impression that Germany was the war's principal instigator. Had Tuchman consulted secondary works available to her by scholars such as Luigi Albertini, she would not have neglected the influence of events in Eastern Europe on Germany's actions.

Text 2

Barbara Tuchman's *The Guns of August* is an engrossing if dated introduction to World War I. Tuchman's analysis of primary documents is laudable, but her main thesis that European powers committed themselves to a catastrophic outcome by refusing to deviate from military plans developed prior to the conflict is implausibly reductive.

Which choice best describes a difference in how the authors of Text 1 and Text 2 view Barbara Tuchman's *The Guns of August*?

- A. The author of Text 1 argues that Tuchman should have relied more on the work of other historians, while the author of Text 2 implies that Tuchman's most interesting claims result from her original research.
- B. The author of Text 1 believes that the scope of Tuchman's research led her to an incorrect interpretation, while the author of Text 2 believes that Tuchman's central argument is overly simplistic.
- C. The author of Text 1 asserts that the writing style of *The Guns of August* makes it worthwhile to read despite any perceived deficiency in Tuchman's research, while the author of Text 2 focuses exclusively on the weakness of Tuchman's interpretation of events.

- D.** The author of Text 1 claims that Tuchman would agree that World War I was largely due to events in Eastern Europe, while the author of Text 2 maintains that Tuchman would say that Eastern European leaders were not committed to military plans in the same way that other leaders were.

Text 1

Dominique Potvin and colleagues captured five Australian magpies (*Gymnorhina tibicen*) to test a new design for attaching tracking devices to birds. As the researchers fitted each magpie with a tracker attached by a small harness, they noticed some magpies without trackers pecking at another magpie's tracker until it broke off. The researchers suggest that this behavior could be evidence of magpies attempting to help another magpie without benefiting themselves.

Text 2

It can be tempting to think that animals are deliberately providing help when we see them removing trackers and other equipment from one another, especially when a species is known to exhibit other cooperative behaviors. At the same time, it can be difficult to exclude the possibility that individuals are simply interested in the equipment because of its novelty, curiously pawing or pecking at it until it detaches.

Based on the texts, how would the author of Text 2 most likely respond to the researchers' perspective in Text 1 on the behavior of the magpies without trackers?

- A. That behavior might have been due to the novelty of the magpies' captive setting rather than to the novelty of the tracker.
- B. That behavior likely indicates that the magpies were deliberately attempting to benefit themselves by obtaining the tracker.
- C. That behavior may not be evidence of selflessness in *Gymnorhina tibicen* because not all the captured magpies demonstrated it.
- D. That behavior might be adequately explained without suggesting that the magpies were attempting to assist the other magpie.

Text 1

Mycoprotein is a fungal biomass that can be eaten as an alternative to meat. Studies of the environmental impact of its manufacture generally agree it is lower than that of beef and closer to that of chicken or pork. But the expense of producing mycoprotein restricts its availability to a few countries with postindustrial economies. Knowing that cost reductions would expand access to mycoprotein, biochemists are exploring solutions, such as a cheaper substrate to feed the mycoprotein as it grows.

Text 2

Cattle farming is a principal cause of global deforestation, and a study by Florian Humpenöder and his colleagues found that replacing 20% of beef consumption worldwide with consumption of mycoprotein would cut deforestation by half if accomplished over the next thirty years. However, this would likely involve only a small change in agricultural water consumption, since water once dedicated to raising cattle would be diverted to raising crops instead.

Based on the texts, how would the author of Text 1 most likely respond to the study findings mentioned in Text 2?

- A. By emphasizing that since agricultural water consumption would remain static in the event of replacing beef consumption with mycoprotein consumption, an effort must be made to substitute mycoprotein for chicken and pork in diets as well
- B. By asserting that the development of a more inexpensive substrate for mycoprotein production would contribute to the goal of decreasing worldwide deforestation over time
- C. By noting that most people would be more likely to use mycoprotein as a substitute for chicken or pork in their diets than as a substitute for beef

- D. By pointing out that some countries are responsible for greater deforestation than others and thus, to have any significant effect on the environment, will have to replace more than 20% of their beef consumption with mycoprotein

Text 1

Good art often challenges and disrupts social and aesthetic norms, but the creation of public art—paintings, sculptures, and performance pieces displayed in nonmuseum or nontheatrical public settings—typically requires broad agreement among artists, civic officials, and community members about the works’ message and artistic goals. Public art that fails to appease everyone by being sufficiently aesthetically and conceptually bland almost inevitably provokes backlash.

Text 2

Public art is commonly displayed in spaces intended for purposes other than meaningful aesthetic engagement. Some critics of public art therefore note that norm-defying pieces that aren’t effectively integrated within their surroundings in a manner that primes passersby to appreciate the pieces’ merits (as is often the case) tend to be regarded more unfavorably than similarly provocative art encountered in museums is.

Based on the texts, how would the critics mentioned in Text 2 most likely respond to the underlined claim in Text 1?

- A. By arguing that the reason members of the general public might disagree about a public artwork’s merits is unrelated to the unconventionality of its appearance and ideas
- B. By agreeing with the idea that only works of art that are universally appealing are suitable for displaying in public spaces
- C. By disputing the notion that civic leaders and community members are easily placated by art that is intended mainly to reinforce social norms

- D. By contending that the kinds of reactions controversial public artworks often receive aren't exclusively the result of attributes inherent in the works themselves

Text 1

Most animals can regenerate some parts of their bodies, such as skin. But when a three-banded panther worm is cut into three pieces, each piece grows into a new worm. Researchers are investigating this feat partly to learn more about humans' comparatively limited abilities to regenerate, and they're making exciting progress. An especially promising discovery is that both humans and panther worms have a gene for early growth response (EGR) linked to regeneration.

Text 2

When Mansi Srivastava and her team reported that panther worms, like humans, possess a gene for EGR, it caused excitement. However, as the team pointed out, the gene likely functions very differently in humans than it does in panther worms. Srivastava has likened EGR to a switch that activates other genes involved in regeneration in panther worms, but how this switch operates in humans remains unclear.

Based on the texts, what would the author of Text 2 most likely say about Text 1's characterization of the discovery involving EGR?

- A. It is reasonable given that Srivastava and her team have identified how EGR functions in both humans and panther worms.
- B. It is overly optimistic given additional observations from Srivastava and her team.
- C. It is unexpected given that Srivastava and her team's findings were generally met with enthusiasm.
- D. It is unfairly dismissive given the progress that Srivastava and her team have reported.

Text 1

Chemically analyzing bulk rock-core samples from Australia's Mount McRae Shale, Ariel Anbar et al. found a transient but significant increase in molybdenum (Mo) at a point corresponding to roughly 2.5 billion years ago (Ga). On Earth, Mo is released mainly through oxidative weathering of minerals; Anbar et al. therefore concluded that atmospheric oxygen briefly increased 2.5 Ga, then returned to its earlier negligible level.

Text 2

Sarah Slotznick et al. reexamined the Mount McRae Shale core. Since chemically analyzing bulk samples can occlude contextual details, Slotznick et al. also employed high-resolution microscopy, which revealed volcanic debris—a known Mo host—around 2.5 Ga and microfractures in the surrounding matrix. The researchers assert that fluid could have reached the debris through the microfractures and initiated oxidative weathering long after debris deposition.

Based on the texts, Anbar et al. (Text 1) and Slotznick et al. (Text 2) would most likely disagree about the answer to which question about the portion of the Mount McRae Shale rock-core corresponding to 2.5 Ga?

- A. Does the increase in Mo in that portion suggest that atmospheric oxygen levels were not negligible before 2.5 Ga?
- B. Is the increase in Mo in that portion indicative of an increase in atmospheric oxygen dating to the same time?
- C. Is the increase in Mo in that portion attributable to oxidative weathering of the material found in that portion?

- D. Did chemically analyzing bulk samples lead to a false impression that there is an increase in Mo in that portion?

Text 1

Ecologists have long wondered how thousands of microscopic phytoplankton species can live together near ocean surfaces competing for the same resources. According to conventional wisdom, one species should emerge after outcompeting the rest. So why do so many species remain? Ecologists' many efforts to explain this phenomenon still haven't uncovered a satisfactory explanation.

Text 2

Ecologist Michael Behrenfeld and colleagues have connected phytoplankton's diversity to their microscopic size. Because these organisms are so tiny, they are spaced relatively far apart from each other in ocean water and, moreover, experience that water as a relatively dense substance. This in turn makes it hard for them to move around and interact with one another. Therefore, says Behrenfeld's team, direct competition among phytoplankton probably happens much less than previously thought.

Based on the texts, how would Behrenfeld and colleagues (Text 2) most likely respond to the "conventional wisdom" discussed in Text 1?

- A. By arguing that it is based on a misconception about phytoplankton species competing with one another
- B. By asserting that it fails to recognize that routine replenishment of ocean nutrients prevents competition between phytoplankton species
- C. By suggesting that their own findings help clarify how phytoplankton species are able to compete with larger organisms

- D. By recommending that more ecologists focus their research on how competition among phytoplankton species is increased with water density

Text 1

Virginia Woolf's 1928 novel *Orlando* is an oddity within her body of work. Her other major novels consist mainly of scenes of everyday life and describe their characters' interior states in great detail, whereas *Orlando* propels itself through a series of fantastical events and considers its characters' psychology more superficially. Woolf herself sometimes regarded the novel as a minor work, even admitting once that she "began it as a joke."

Text 2

Like Woolf's other great novels, *Orlando* portrays how people's memories inform their experience of the present. Like those works, it examines how people navigate social interactions shaped by gender and social class. Though it is lighter in tone—more entertaining, even—this literary "joke" nonetheless engages seriously with the themes that motivated the four or five other novels by Woolf that have achieved the status of literary classics.

Based on the texts, how would the author of Text 2 most likely respond to the assessment of *Orlando* presented in Text 1?

- A. By conceding that Woolf's talents were best suited to serious novels but asserting that the humor in *Orlando* is often effective
- B. By agreeing that *Orlando* is less impressive than certain other novels by Woolf but arguing that it should still be regarded as a classic
- C. By acknowledging that *Orlando* clearly differs from Woolf's other major novels but insisting on its centrality to her body of work nonetheless
- D. By concurring that the reputation of *Orlando* as a minor work has led readers to overlook this novel but maintaining that the reputation is unearned

Text 1

The Cretaceous-Paleogene (K-Pg) mass extinction event is usually attributed solely to an asteroid impact near Chicxulub, Mexico. Some scientists argue that volcanic activity was the true cause, as the K-Pg event occurred relatively early in a long period of eruption of the Deccan Traps range that initially produced huge amounts of climate-altering gases. These dissenters note that other mass extinctions have coincided with large volcanic eruptions, while only the K-Pg event lines up with an asteroid strike.

Text 2

In a 2020 study, Pincelli Hull and her colleagues analyzed ocean core samples and modeled climate changes around the K-Pg event. The team concluded that Deccan Traps gases did affect global conditions prior to the event, but that the climate returned to normal well before the extinctions began—extinctions that instead closely align with the Chicxulub impact.

Based on the texts, how would Hull’s team (Text 2) most likely respond to the argument in the underlined portion of Text 1?

- A. By agreeing that the Chicxulub impact changed the climate and that the Deccan Traps eruption caused the K-Pg event
- B. By declaring that the changes in climate caused by the Deccan Traps eruption weren’t the main cause of the K-Pg event
- C. By questioning why those scientists assume that the Chicxulub impact caused the Deccan Traps eruption
- D. By asserting that the Deccan Traps eruption had a more significant effect on global conditions than those scientists claim